

1 From reciprocal-space proof to benchmark identity ★ evidence, not table transcription

C1 + C2 · VERIFIED

7,680
rotary identities

7,936
symmetry trials

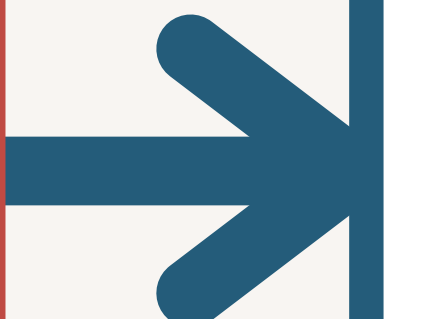
4.46e-14
worst C1 math error



C5 · FALSIFIED

ShapeNetPart
≠
ModelNet40

released path: 16 categories
50 part labels · part mIoU
0 SemanticKITTI files



C3 + C4 + C6
INCONCLUSIVE

0
released checkpoints

396 vs 384
nD vs baseline width

exact metrics require trained models

exact certificate → executable source audit → honest replication boundary

Reproduction evidence map. Exact Fourier and simplex certificates resolve C1/C2; executable dataset paths falsify C5 as written; missing checkpoints bound C3/C4/C6.

The clean-room checks instantiate the paper's equations directly and then execute the pinned author functions for parity. The empirical audit scans all 154 released files, full Git history, Hub assets, the only fork, and local caches without promoting reported tables to reproduced measurements.

2 C1 — Fourier/Hilbert derivation

A finite Fourier analogue checks Parseval transfer, kernel preservation, multiplicative phase modulation, and displacement-only attention. The pinned author implementation is evaluated verbatim through an AST-isolated parity harness.

- 7,680 rotary translation/displacement identities.
- 192 finite Parseval, relative-shift, and Riesz cases.
- Worst mathematical error: 4.46×10^{-14} .

3 C2 — simplex geometry and scale

The regular simplex is checked from dimension 2 through 32 for rank, centroid, Gram geometry, tight-frame energy, and permutation symmetry. A separate scalar optimizer tests the economy principle.

ECONOMY OPTIMUM

$$\rho = r^n = e \implies r^* = e^{1/n}$$

7,936 symmetry trials, 64 independent optimizations, and 15/15 malformed-geometry controls pass.

Certificate	Cases	Worst error
Eq. 18 Gram	31	9.75e-16
Eq. 19 frame	31	1.55e-15
Scale optimum	64	4.44e-8

4 C5 — benchmark identity falsified

The anchored claim calls 85.97 mIoU a ModelNet40 segmentation result. Both released training and test entrypoints instead load ShapeNetPart, use 16 object categories and 50 part labels, and calculate part IoU. The separate ModelNet40 path performs 40-way **classification**; it cannot produce the reported instance-average part mIoU. The release also has zero SemanticKITTI files.

Decisive result. ShapeNetPart ≠ ModelNet40. One false benchmark identity falsifies this conjunctive claim as written; Kinetics and SemanticKITTI remain untested.

5 Unresolved empirics and audit record

Apple M2 CPU; pinned source; no checkpoint, dataset, paid compute, or hidden table substitution.

20/20
tests
passing

154
source
files audited

0
released
checkpoints

C3 and C4 need the unreleased trained ImageNet models. Empirical C6 needs ablation checkpoints or full retraining; table arithmetic alone is not reproduction.

Image width: 396 vs 384; Table 8: +6% FLOPs and parameters.

Four A100s cost \$10/hour; no paid job was launched.

6 Reproduction ledger | FOURIER. Translation identity and Parseval chain verify.

| GEOMETRY. Simplex symmetry and $e^{1/n}$ optimum verify.

| IDENTITY. ShapeNetPart is not ModelNet40.

| LIMITS. No checkpoint means no exact accuracy claim.